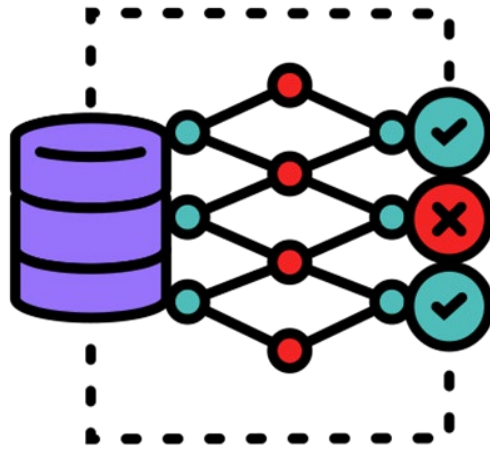




DECISION SUPPORT SYSTEMS

2025

PRACTICAL & ORAL BANK

(Solved)




SENIOR
25



ORAL BANK

Choose the correct answer:

1.	an interrelated set of components including people, activities, technology, and procedures that are designed or intended to achieve a predefined purpose			
	a) system	b) Goal	c) Resources	d) None
2.	is a process by which organizational goals are achieved through the use of resources.			
	a) system	b) Management	c) Resources	d) Goal
3.	Resources refer to.....			
	a) Inputs	b) Output	c) Outputs / Inputs	d) None
4.	Goal Attainment refer to.....			
	a) Inputs	b) Output	c) Outputs / Inputs	d) None
5.	Measuring Success refer to			
	a) Inputs	b) Output	c) Outputs / Inputs	d) None
6.	 awareness or familiarity gained by experience of a fact or situation .			
	a) Knowledge	b) Information	c) Data	d) None
7.	 comprises things that are held to be true in each context and that drive us to action if there were no impediments			
	a) Knowledge	b) Information	c) Data	d) None
8.	 defined as processed data is an increment in knowledge : it contributes to the general framework of concepts and facts that we know.			
	a) Knowledge	b) Information	c) Data	d) None
9.	 are only the raw facts, the material for obtaining information . <u>Information systems use data stored in computer databases to provide needed information.</u>			
	a) Knowledge	b) Information	c) Data	d) None
10.	 an organized collection of interrelated data reflecting a major aspect of a firm's activities			
	a) Database	b) DBMS	c) Downsizing	d) None
11.	 is an elementary activity conducted during business Operations.			
	a) Transaction	b) system	c) Modelling	d) None
12.	TPS may work either inmode, processing accumulated transactions at a single time later on or inmode, processing incoming transactions immediately.			
	a) batch, on-line	b) on-line, batch	c) system, Modelling	d) None
13.	TPS today generally work inmode by immediately processing a firm's business transactions.			
	a) on-line	b) batch	c) Transaction	d) None



14.	 The objective of management reporting systems is to provide routine information to managers.							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
15.	 Managers receive performance reports within their specific areas of responsibility							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
16.	 Generally, these reports provide internal information rather than spanning corporate boundaries.							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
17.	 They report on the past and the present, rather than projecting the future							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
18.	 These systems facilitate a dialog between the user , who is considering alternative problem solutions , and the system that provides built-in models and access to databases .							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
19.	Thedatabases are often extracted from the general databases of the enterprise or from external databases .							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
20.	 support top managers with conveniently displayed summarized information, customized for them.							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
21.	Executive Information Systems are use to							
	a)	Monitor the performance of the organization.	b)	Assess the business environment.	c)	Develop strategic directions for the company's future.	d)	All the above
22.	is to facilitate communication between the members of an organization and between the organization and its environment							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Office Information Systems (OIS)
23.	 are system that employs knowledge about its application domain and uses an inferencing (reason) procedure to solve problems that would otherwise require human competence or expertise .							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Expert systems (ES)



24.	 Process of choosing amongst alternative courses of action for the purpose of attaining a goal or goals.							
	a)	Decision Making	b)	Decision	c)	Intelligence	d)	None
25.	 is a cognitive process that results in the selection of a course of action among several alternative scenarios.							
	a)	Decision Making	b)	Decision	c)	Intelligence	d)	None
26.	It contains data from various sources, including internal data from the organization, the data generated by different applications, and the external data mined from the Internet							
	a)	DSS Database	b)	DSS Software System	c)	DSS User Interface	d)	None
27.	It consists of various mathematical and analytical that are used to analyze complex data, thereby producing the required information.							
	a)	DSS Database (DSS DB)	b)	DSS Software System (DSS SW)	c)	DSS User Interface (DSS UI)	d)	DSS Dialogue
28.	 It is an interactive graphical interface which makes the interaction easier between the DSS and its users. It displays the results (output) of the analysis in various forms, such as text, table, charts or graphics							
	a)	DSS Database	b)	DSS Software System	c)	DSS User Interface (DSS UI)	d)	None
29.	 Supports decision making by analyzing given time-series of data and returning new information gained by those analysis. or <u>It is used to query a database or data warehouse to seek specific answers for specific purposes.</u>							
	a)	data-driven DSS	b)	document-driven DSS	c)	Knowledge-driven DSS	d)	Model- driven DSS
30.	The purpose of such a DSS is to search web pages and find documents on a specific set of keywords or search terms . Or <u>the usual technology used to set up such DSSs are via the web or a client/server system</u>							
	a)	data-driven DSS	b)	document-driven DSS	c)	Knowledge-driven DSS	d)	Model- driven DSS
31.	 The typical deployment technology used to set up such systems could be client/server systems, the web, or software running on stand-alone PCs. Or <u>It is essentially used to provide management advice or to choose products/services.</u>							
	a)	data-driven DSS	b)	document-driven DSS	c)	Knowledge-driven DSS	d)	Model- driven DSS
32.	 These are used by managers and staff members of a business, or people who interact with the organization, for several purposes depending on how the model is set up - scheduling, decision analyses							
	a)	data-driven DSS	b)	document-driven DSS	c)	Knowledge-driven DSS	d)	Model-driven DSS
33.	 is a technology-driven process for analyzing data and delivering actionable information that helps executives, managers and workers make informed business decisions.							
	a)	BI	b)	DSS	c)	ERD	d)	None
34.	 Repetitive and routine , and decision makers can follow a definite procedure for handling them to be efficient							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None



35.	 Only part of the problem has a clear-cut answer provided by an accepted procedure.							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
36.	Each of these decisions is novel, important, and nonroutine , and there is no well-understood or agreed-on procedure for making them							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
37.	Decisions are those in which the decision maker must provide judgment , evaluation, and insights into the problem definition.							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
38.	It is assumed that complete knowledge is available so that the decision-maker knows exactly what the outcome of each course of action will be (as in a deterministic environment) example for decision making Under.....							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
39.	The decision-maker is viewed as a perfect predictor of the future because it is assumed that there is only <i>one outcome</i> for each alternative example for decision making Under							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
40.	Every time you park downtown, you get a parking ticket because you exceed the time limit on the parking meter <i>although once it did not happen</i> , this is an example for decision making under							
	(بتروح تركن بعرييتك ولو قعدت اكثر من المده المتاحه بتاخذ غرامه. وكل مره بتركن هناك وتعدي مدتلك المتاحه بس في مره ركنت ربع ساعة بس وبرضه خدت غرامه عشان هو متأكد انه انت اللي كل مره بتركن كثير)							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
41.	The decision-maker considers situations in which several outcomes are possible for each course of action. <u>In this case the decision-maker does not know, or cannot estimate, the probability of occurrence of the possible outcomes</u> example for decision making under.....							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
42.	The decision-maker must consider several possible outcomes for each alternative, each with a given probability of occurrence under.....							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
43.	The output of Traditional DSS are decisions based on data							
	a)	Quantitative	b)	Qualitative	c)	Both A&B	d)	None
44.	The output of Knowledge-based DSS are decisions based on..... data							
	a)	Quantitative	b)	Qualitative	c)	Both A&B	d)	None
45.	phase of DSS process consists of choosing among solution alternatives.							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation
46.	Sensitivity analysis is applied in which phase of DSS process.							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation
47.	 phase of DSS process consists of discovering, identifying, and understanding the <u>problems occurring</u> in the organization.							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation



48.	 phase of DSS process involves identifying and exploring various <u>solutions</u> to the problem							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation
49.	 involves identifying and exploring various <u>solutions</u> to the problem.							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation
50.	 phase of DSS process involves making the chosen alternative work and continuing to monitor how well the solution is working							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation
51.	Model are used to provide answers to what-if situations occurring frequently in an organization							
	a)	Sensitivity Analysis	b)	Statistical	c)	Forecasting	d)	Backward Analysis
52.	 Model are known as goal seeking analysis, the technique followed in these models is just opposite to the technique applied in sensitivity analysis models.							
	a)	Sensitivity Analysis	b)	Statistical	c)	Forecasting	d)	Backward Analysis
53.	Model are contain a wide range of statistical functions, such as mean, median, mode , deviations etc. <u>These models are used to establish, relationships between the occurrences of an event and various factors related to that event</u>							
	a)	Sensitivity Analysis	b)	Statistical	c)	Forecasting	d)	Backward Analysis
54.	 Model are used to find optimum value for a target variable under given circumstances.							
	a)	Sensitivity Analysis	b)	Optimization Analysis	c)	Forecasting	d)	Backward Analysis
55.	 Model are use various forecasting tools and techniques.							
	a)	Sensitivity Analysis	b)	Optimization Analysis	c)	Forecasting	d)	Backward Analysis
56.	may work either in batch mode, processing accumulated transactions at a single time later on, or in on-line mode, processing incoming transactions immediately.							
	a)	TPS	b)	MIS	c)	DSS	d)	EIS
57.	support top managers with conveniently displayed summarized information, customized for them.							
	a)	TPS	b)	MIS	c)	DSS	d)	EIS
58.	The main objective of..... is to facilitate communication between the members of an <i>organization</i> and between the organization and its environment.							
	a)	TPS	b)	MIS	c)	DSS	d)	OIS
59.	 These systems facilitate a dialog between the user , who is considering <i>alternative problem solutions</i> , and the system that provides built-in models and access to databases.							
	a)	TPS	b)	MIS	c)	DSS	d)	EIS
60.	is the increase in the expected profit that would result if one knew with certainty which state of nature would occur							
	a)	AHP	b)	EV	c)	ROMC	d)	EVPI
61.	If probabilistic information regarding the states of nature is available, one may use the.....							
	a)	AHP	b)	EV	c)	ROMC	d)	EVPI



62.	 is done by comparing model's output and the actual behavior of the event .							
	a)	Model validation	b)	Modeling	c)	Assumptions	d)	Forecasts
63.	 is the process of identifying an appropriate model for a prospective model-driven decision support system, Or <u>is done, it's vital to validate the selected model, to ensure it works well and generates appropriate results.</u>							
	a)	Model validation	b)	Modeling	c)	Assumptions	d)	Forecasts
64.	Analysis of take into consideration the long-term response of a system. Or <u>It takes a single snapshot of a situation</u> and assumes that it will remain stable all through and won't change.							
	a)	Static	b)	Dynamic	c)	Explanatory	d)	none
65.	 Analysis is testing a program or a software system in real-time . Or <u>This method considers that the situation changes over time, due to any reason, such as cost, rules and regulations, time, etc.</u>							
	a)	Static	b)	Dynamic	c)	Explanatory	d)	none
66.	Model Describes and explains why something is the way it is and why and how it works.							
	a)	Descriptive (Explanatory)	b)	Contemplative	c)	Algebraic	d)	none
67.	 Model Forecasts results or outcomes that may be produced from a specific set of parameters.							
	a)	Descriptive (Explanatory)	b)	Contemplative	c)	Algebraic	d)	none
68.	 Model high-level modeling system for solving complex equations . It is employed to optimize a variable or equation. The best part is that it can handle several simultaneous equations.							
	a)	Descriptive (Explanatory)	b)	Contemplative	c)	Algebraic	d)	none
69.	 summarize the anticipated financial results for a specific period in the future.							
	a)	Pro Forma	b)	Break-Even	c)	AHP model	d)	Ratio
70.	 a financial tool that helps businesses determine the <u>sales</u> volume or revenue needed to cover all costs associated with producing and selling a product or service.							
	a)	Pro Forma	b)	Break-Even	c)	AHP model	d)	Ratio
71.	a multi-criteria decision technique (MCDM) that combines quantitative and qualitative factors when evaluating alternatives.							
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)	d)	Payoff Table (PT)
72.	 emphasizes the quantification of individual preferences using utility functions .							
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)	d)	Payoff Table (PT)
73.	The consequence resulting from a specific combination of a decision alternative and a state of nature is a							
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)	d)	Payoff



74.	A is a graphical representation of the decision problem.							
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)	d)	Payoff Table (PT)
75.	Each decision tree has two types of nodes; round nodes correspond to while square nodes correspond to							
	a)	Decision alternatives, States of nature	b)	States of nature, Decision alternatives	c)	Utility functions, multi-criteria	d)	Multi-criteria, Utility functions
76.	 Choose decision with the maximum of the maximum payoffs.							
	a)	Maximax (optimistic)	b)	Maximin (pessimistic)	c)	Minimax (regret)	d)	None
77.	 Choose decision with the maximum of the minimum payoffs.							
	a)	Maximax (optimistic)	b)	Maximin (pessimistic)	c)	Minimax (regret)	d)	None
78.	 Choose decision with the minimum of the maximum regrets for each alternative.							
	a)	Maximax (optimistic)	b)	Maximin (pessimistic)	c)	Minimax (regret)	d)	None
79.	MAUA is considered.....tool							
	a)	Financial	b)	budget analysis	c)	decision analysis	d)	optimization
80.	 <u>simple</u> forecasting that provides limited accuracy							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Regression	d)	Exponential Smoothing
81.	 the perceptions and opinions of experts instead on hard data, long term forecasting.							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Regression	d)	Exponential Smoothing
82.	 the predictions are based upon the historical values .							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Moving Average	d)	Exponential Smoothing
83.	alters the historical data mathematically to better reflect the assumptions of a decision maker							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Moving Average	d)	Exponential Smoothing
84.	 considers the economic variables that are measured at consecutive intervals of time . Or <u>It is believed that the knowledge of past behavior of the variable at successive intervals of time will help understand the behavior of the variables in future better.</u>							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Moving Average	d)	Time Series Extrapolation
85.	 use of linear and multiple regressions to establish <i>cause and effect</i> relationships							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Regression & Econometric Models	d)	Exponential Smoothing



86.	 conduct experiments to identify conditions or situations that approximate the actual conditions.							
	a)	simulation model	b)	Network and optimization models	c)	Forecasting model	d)	none
87.	 All models are made up of three basic components : decision variables, uncontrollable variables (and/or parameters), and result (outcome) variables.							
	a)	Mathematical models	b)	Network and optimization models	c)	Forecasting model	d)	none
88.	is a model-driven DSS to help a person decide how much he or she needs to <u>invest</u> each month <u>for retirement</u> .							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
89.	 a web-based system that allows users to make pair-wise comparisons of car models.							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
90.	a knowledge-driven web-based decision support systems, offering 25 different decision guides on <u>diverse topics</u> , including choosing pets, bikes and business schools.							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Netscape Decision Guides	d)	TCB Works
91.	 It's a data-driven DSS to help investors identify stocks based on various criteria, including industry type, price and earnings							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
92.	 A communication-driven web-based DSS, it enables people to interact, discuss and make both structured and multi-criteria decisions							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
93.	 is a collection of related data and <u>data is a collection of facts and figures</u> that can be processed to produce information							
	a)	Database	b)	DBMS	c)	Downsizing	d)	None
94.	 stores data in such a way that it becomes <u>easier</u> to retrieve, manipulate, and produce information.							
	a)	Database	b)	DBMS	c)	Downsizing	d)	None
95.	 is an organized approach to design specialized decision support systems , more precisely their user interfaces. Or <u>is a systematic move towards designing the representation, operation, memory and control mechanisms of a large decision support system.</u>							
	a)	MAUA	b)	AHP	c)	GUI	d)	ROMC
96.	 is about presenting information or results for that matter, in a structured way. Or <u>provides a base to the decision makers backed by concrete information, helping them interpret DSS outputs.</u> And <u>can be in the form of a table, graph, map, chart or a text document</u> and each value on a map or a table <u>communicates decision making context.</u>							
	a)	Representation	b)	operation	c)	memory aid	d)	none
97.	is stage in user interface design focuses on specific tasks performed by/with a decision support system. Or <u>an involve one activity or many, depending upon the specific needs of the decision maker.</u>							
	a)	Representation	b)	operation	c)	memory aid	d)	none



98.	 measures the extent to which the system meets its objectives .							
	a)	reliability	b)	effectiveness	c)	efficiency	d)	completeness
99.	DSS support all organization levels except level.							
	a)	strategic	b)	operational	c)	management	d)	tactical
100.	The decision-making process can be defined as the activity that leads to the solution of a decision-making problem involving at least Alternatives							
	a)	one	b)	two	c)	three	d)	four
101.	Data availability is more in which of the following?							
	a)	strategic	b)	operational	c)	management	d)	tactical
102.	 improving communication and coordination among people with organizations to improve organization performance and solve problems							
	a)	web-based DSS	b)	GDSS	c)	KBDSS	d)	ODSS
103.	The output of the DSS design phase is							
	a)	alternatives	b)	Problem statement	c)	criteria	d)	solutions
104.	If a company has fixed costs of \$100,000, variable costs of \$10 per unit, and a selling price of \$20 per unit, This means that the company must sell units to cover all of its costs. <i>(Break-even = Fixed cost / (selling price – variable cost)</i> <i>= 100000 / (20 – 10))</i>							
	a)	1000	b)	2000	c)	5000	d)	10000
105.	 analysis can help you to compare your company's performance to that of other companies in your industry .							
	a)	Pro Forma	b)	Break-Even	c)	AHP model	d)	Ratio
106.	is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies.							
	a)	DSS	b)	BI	c)	Data mining	d)	MSS
107.	The process of is based on the transformation of data to information, then to decisions, and finally to actions .							
	a)	DSS	b)	BI	c)	Data mining	d)	MSS
108.	Managers and specialists use the <u>concept of a system</u> frequently and yet it is hard for most of us to define and understand the concept.							
	a)	EIS	b)	TPS	c)	MIS	d)	DSS
109.	Systems also have mechanisms to provide a means of controlling the operation of the system.							
	a)	input	b)	output	c)	feedback	d)	results
110.	Depending on the type of, the information output may take the form a <u>query response, decision outcome, expert-system advice, transaction document, or a report</u> .							
	a)	DSS	b)	MIS	c)	IS	d)	ES
111.	The objectives of a system are realized in its							
	a)	input	b)	output	c)	feedback	d)	results



112.	 measuring performance against the planned objectives and <i>initiating corrective action</i> , if needed			
	a) planning	b) leadership	c) controlling	d) organizing
113.	 <i>establishing goals</i> and selecting the actions needed to achieve them over a specific period of time			
	a) planning	b) leadership	c) controlling	d) organizing
114.	process of establishing worker relationships allows workers to work together to achieve their organizational goals .			
	a) planning	b) leadership	c) controlling	d) organizing
115.	including the people in the organization to contribute to its goals			
	a) planning	b) leadership	c) controlling	d) organizing
116.	 Recruiting and selecting employees for positions within the company (within teams and departments)			
	a) planning	b) leadership	c) Staffing	d) organizing
117.	clarifies the computer's role in aiding rather than replacing the decisionmaker.			
	a) Decision	b) support	c) systems	d) DSS
118.	Its purpose is to help conduct a meeting, or for users to collaborate driven DSS.			
	a) Office	b) Data	c) Communication	d) document
119.	is an approach to discovery and problem solving by employing practical methods.			
	a) Heuristics	b) Inference engine	c) Knowledge acquisition	d) Case reasoning
120.	 used to determine the <u>distance between or relationship among various components</u> .			
	a) Heuristics	b) Inference engine	c) Knowledge acquisition	d) Case reasoning
121.	“What techniques will it use?” used in for designing web-based DSS.			
	a) conceptualization	b) Problem identification	c) Feasibility analysis	d) SW Development
122.	 means that a decision support system must be resourceful enough to perform the entire range of tasks that a decision maker needs to perform when making a decision using DSS.			
	a) Adaptability	b) Versatility	c) Quality	d) Fatigue
123.	 helps you to identify <u>all possible causes</u> of the problem.			
	a) Cause-and-Effect diagram	b) Pareto Chart	c) Decision tree	d) None
124.	 helps you to prioritize and identify the causes <u>with the highest effect</u> .			
	a) Cause-and-Effect diagram	b) Pareto Chart	c) Decision tree	d) None
125.	The _____ phase involves defining the purpose and objectives of the DSS and identify the specific decision-making problems or opportunities that the DSS will address			
	a) Initialization	b) Decision Oriented Diagonal	c) Feasibility study	d) None



126.	The _____ to knowledge management attempts to codify organizational knowledge through formalized controls, processes, and technologies				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
127.	Organizations that adopt the _____ may implement explicit policies governing how knowledge is to be collected, stored, and disseminated throughout the organization				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
128.	_____ the knowledge may be typically static in nature				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
129.	The _____ to knowledge management assumes that a great deal of organizational knowledge is tacit in nature and that formal controls, processes, and technologies are not suitable for transmitting this type of understanding				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
130.	The focus of _____ approach is to build the social environments or communities of practice necessary to facilitate the sharing of tacit understanding				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
131.	_____ is typically adopted by companies that provide highly customized solutions to unique problems.				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
132.	Many organizations use a _____				
	a)	process approach	b)	Practice approach	c) Hybrid approach d) None
133.	_____ is any combination of hardware and software that enhances groupwork either in direct or indirect support of decision making				
	a)	GDSS (group decision support system)	b)	GSS (group support system)	c) DSS (decision support system) d) DGSS
134.	_____ an interactive computer-based system that facilitates the solution of semi structured or unstructured problems <i>by a group of decision makers.</i>				
	a)	GDSS	b)	GSS	c) DSS d) DGSS
135.	_____ is the systematic and active management of ideas, information, and knowledge residing in an organization's employees				
	a)	Knowledge management (KM)	b)	knowledge process (KP)	c) Decision Support System (DSS) d) All are true
136.	_____ process that helps organizations identify, select, organize, disseminate, and transfer important information and expertise that are part of the organization's memory and that typically reside within the organization in an unstructured manner.				
	a)	KM (knowledge management)	b)	KP (knowledge process)	c) DSS (Decision Support System) d) All are true
137.	_____ is a family of methods that attempt to allow machines to acquire knowledge for problem-solving by showing them historical cases				
	a)	Machine learning	b)	supervised learning	c) Unsupervised Learning d) Reinforcement Learning
138.	_____ is used for making predictions from data . To be able to do that, we need to know what to predict, which is also known as the target variable				
	a)	Machine learning	b)	supervised learning	c) Unsupervised Learning d) Reinforcement Learning



139.	_____ is the science of decision making. It is about learning the optimal behaviour in an environment to obtain maximum reward .							
	a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning
140.	_____ is self-organized learning. Its main aim is to explore the underlying patterns and predicts the output							
	a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning
141.	_____ is a problem-solving methodology that attempts to mimic how our <u>brains</u> function.							
	a)	Machine learning	b)	Neural computing	c)	Unsupervised Learning	d)	Reinforcement Learning
142.	The human brain is composed of special cells called _____							
	a)	Neurons	b)	Axon hillock	c)	Cell body	d)	Axon
143.	In Biological and artificial neural, Dendrites = _____, and Axon = _____							
	a)	Input/Input	b)	Input/Output	c)	Output/input	d)	output/ output
144.	In Artificial neural networks, The basic processing unit is the _____							
	a)	Neurons	b)	Axon hillock	c)	Cell body	d)	Axon
145.	A _____ is composed of processing elements organized in different ways to form the network's structure.							
	a)	Neural network	b)	Neural computing	c)	Artificial Neurons	d)	None
146.	The processing elements (PEs) of an ANN are _____							
	a)	Neural network	b)	Neural computing	c)	Artificial Neurons	d)	None
147.	Decision at the top level tend to be more _____							
	a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning
148.	Each ANN is composed of a collection of neurons grouped in three layers _____.							
	a)	input	b)	Intermediate	c)	Output	d)	All
149.	Each _____ corresponds to a single attribute							
	a)	input	b)	Intermediate	c)	Output	d)	All
150.	The _____ of the network contain the solution to a problem							
	a)	input	b)	Intermediate	c)	Output	d)	All
151.	The resulting model from neural computing is often called _____.							
	a)	Neural network	b)	Neural computing	c)	Artificial Neurons Network (ANN)	d)	None
152.	_____ A population of potential solutions (individuals or chromosomes) is <i>randomly generated</i> . Each solution represents a possible solution to the problem at hand.							
	a)	Initialization	b)	Evaluation	c)	Selection	d)	Crossover



153.	_____ Each individual in the population is evaluated for its fitness, which is a measure of how well it solves the problem. The fitness function is problem specific and reflects the objective or goal of the optimization							
	a)	Initialization	b)	Evaluation	c)	Selection	d)	Crossover
154.	_____ Individuals are selected from the population based on their fitness. The more fit individuals have a higher chance of being selected .							
	a)	Initialization	b)	Evaluation	c)	Selection	d)	Crossover
155.	Pairs of selected individuals (parents) <u>exchange</u> genetic information to create new individuals (offspring) in the _____ phase of the genetic algorithm							
	a)	Mutation	b)	Replacement	c)	Termination	d)	Crossover
156.	_____ <i>Random changes</i> are applied to some of the individuals in the population. This introduces new genetic material and helps explore the solution space more broadly							
	a)	Mutation	b)	Replacement	c)	Termination	d)	Crossover
157.	_____ the new offspring and possibly some of the existing individuals are used to form the next generation. The less fit individuals may be replaced by the fitter offspring							
	a)	Mutation	b)	Replacement	c)	Termination	d)	Crossover
158.	_____ the algorithm continues to iterate through the selection, crossover, mutation, and replacement steps for a fixed number of generations or until a termination condition is met.							
	a)	Mutation	b)	Replacement	c)	Termination	d)	Crossover
159.	_____ is a widely applied concept to deal with uncertainty .							
	a)	Fuzzy theory	b)	Fuzzy logic	c)	Both A&B	d)	Fuzzy Sets
160.	_____ concepts that belong to the field of fuzzy set theory , a mathematical framework for dealing with uncertainty and vagueness.							
	a)	Fuzzy theory	b)	Fuzzy logic	c)	Both A&B	d)	Fuzzy Sets
161.	_____ theory allows elements to partially belong to a set .							
	a)	Fuzzy theory	b)	Fuzzy logic	c)	Both A&B	d)	Fuzzy Sets
162.	_____ Repeat the process for a certain number of generations or until a termination condition is met							
	a)	Mutation	b)	Replacement	c)	Termination	d)	Crossover
163.	_____ Replace some individuals in the old population with the new offspring. For simplicity, replace the two least fit individuals.							
	a)	Mutation	b)	Replacement	c)	Termination	d)	Crossover
164.	_____ Select individuals based on their fitness. Higher-performing feature subsets are more likely to be selected.							
	a)	Selection	b)	Replacement	c)	Termination	d)	Crossover
165.	The difference between the actual output (For YT) and the desired output (Z) for a given set of inputs is an error called _____							
	a)	Delta	b)	Sigma	c)	Alpha	d)	None
166.	The _____ function computes the weighted sums of all the input elements entering each processing element.							
	a)	Summation	b)	Transfer	c)	Connection Weights	d)	None



167.	_____ is a broad concept. Its objective is to assess an ES's overall value. In addition to assessing acceptable performance levels, it analyzes whether the system would be usable, efficient, and cost-effective			
	a) Evaluation	b) Validation	c) Verification	d) All
168.	_____ is the part of evaluation that deals with the <u>performance of the system</u> , simply stated, it is building the <u>right system</u>			
	a) Evaluation	b) Validation	c) Verification	d) All
169.	_____ is building the <u>system right</u> or substantiating that the system is correctly implemented to its specifications.			
	a) Evaluation	b) Validation	c) Verification	d) All
170.	_____ declarative rules, state all the facts and relationships about a problem.			
	a) Knowledge rules	b) Inference rules	c) Expect rules	d) None
171.	_____ procedural rules, offer advice on how to solve a problem, given that certain facts are known.			
	a) Knowledge rules	b) Inference rules	c) Expect rules	d) None
172.	_____ is a goal-driven approach in which you start from an expectation of what is going to happen and then seek evidence that supports your expectation. Often, this entails formulating and testing intermediate hypotheses (or sub-hypotheses)			
	a) Forward chaining	b) Backward chaining	c) Reinforcement	d) Supervised DS
173.	_____ is a data-driven approach. We start from available information as it becomes available or from a basic idea, and then we try to draw conclusions.			
	a) Forward chaining	b) Backward chaining	c) Reinforcement	d) Supervised DS
174.	_____ is the process of using the rules in the knowledge base along with the known facts to draw conclusions.			
	a) inferencing	b) reasoning	c) Both A&B	d) None
175.	_____ is represented in the form of condition/action pairs: IF this condition (or premise or antecedent) occurs, THEN some action (or result or conclusion or consequence) will (or should) occur.			
	a) Knowledge	b) inferencing	c) reasoning	d) Production
176.	_____ involves the acquisition of knowledge from human experts, books, documents, sensors, or computer files.			
	a) Knowledge Acquisition	b) Knowledge Validation	c) Knowledge Representation	d) Interfacing
177.	_____ Knowledge acquired from experts or induced from a set of data must be represented in a format that is both understandable by humans and executable on computers.			
	a) Knowledge Acquisition	b) Knowledge Validation	c) Knowledge Representation	d) Interfacing
178.	_____ is the process of using the rules in the knowledge base along with the known facts to draw conclusions .			
	a) Inferencing (reasoning)	b) Validation	c) Evaluation	d) None
179.	_____ is where the information is stored in the expert system in form of facts and rules. Basically, a series of IF statements. Here the programmer writes the code for the expert system.			
	a) knowledge base	b) user interface	c) inference engine	d) None



180.	_____ is where the user interacts with the expert system. This is where questions are asked and advised is produced. It can also output justification features of the expert system			
	a) knowledge base	b) user interface	c) inference engine	d) None
181.	_____ <i>applies the facts to the rules</i> and determines the questions to be asked of the user in the user interface and which order to ask them. This is an invisible part of an ESS when the program is run			
	a) knowledge base	b) user interface	c) inference engine	d) None
182.	During the _____ phase, you conduct a thorough analysis of the current decision-making processes within the organization.			
	a) diagnosis	b) Initialization	c) Make/Buy	d) Feasibility study
183.	_____ is structured approach for evaluating and assessing the effectiveness, efficiency, and security of a Decision Support System within an organization.			
	a) DSS audit plan	b) DSS diagnosis	c) DSS planning	d) DSS evaluation
184.	_____ is a rule-based system that provides a solution, usually in one functional area (e.g., finance, manufacturing), to a specific repetitive managerial problem, usually in one industry.			
	a) ADS	b) ES	c) DSS	d) KBS
185.	_____ developed at Stanford University in the early 1980s for medical diagnosis , is the most well-known ES application			
	a) SONAR	b) LMOS	c) XCON	d) MYCIN
186.	In the influence diagram, _____ represented by hexagons and represent the objectives of the decision-maker.			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) output nodes
187.	In influence diagram, _____ are represented by circles and represent the uncertainties in the problem.			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) output nodes
188.	In influence diagram, _____ are represented by squares and represent the decisions that need to be made			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) output nodes
189.	In influence diagram, _____ represent the relationships between the different elements of the problem.			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) Arcs
190.	_____ is the identification of problems or opportunities for improvement in current decision behavior.			
	a) diagnosis	b) Initialization	c) Make/Buy	d) Feasibility study
191.	_____ is an area of working memory set aside as a database for <u>description of the current problem</u> , as characterized by the input data			
	a) User interface	b) Knowledge base	c) Blackboard	d) Database
192.	The collection of intensive activities encompassing the acquisition of knowledge from human experts and conversion of this knowledge into a repository (commonly called a knowledge base) are called knowledge _____			
	a) acquisition	b) engineering	c) evaluation	d) validation
193.	Random changes are applied to some of the individuals in the population in _____ phase of genetic algorithm.			
	a) Mutation	b) Replacement	c) Termination	d) Crossover



True/False:

1.	Management <u>reporting</u> systems <u>report</u> on the past and the present, rather than projecting the future.	T
2.	Executive information systems support <u>top managers</u> with conveniently displayed summarized information, customized for them.	T
3.	The core management activates for planning in the management level of organization	F
4.	<u>Decision</u> is a <u>choice</u> made between alternative courses of action in a situation of uncertainty.	T
5.	Decision is a <u>cognitive process</u> that results in the selection of a course of action among several alternative scenarios (Decision-making)	F
6.	Pareto chart helps you to identify all possible causes of the problem. (Cause-and-effect)	F
7.	Forward Analysis Sensitivity Models known as <u>goal</u> seeking analysis. (Backward)	F
8.	Decision-making under uncertainty is more difficult because of insufficient information	T
9.	The state of nature are the <u>different possible strategies</u> the decision maker can employ. (decision alternatives)	F
10.	States of nature could be defined as low demand and high demand.	T
11.	Alternatives could decide to build as small, medium, or large condominium complex.	T
12.	Payoffs are the profit for each alternative under each potential state of nature is going to be determined.	T
13.	A model driven DSS may employ a single model or a combination of two or more models.	T
14.	Model-driven DSS are generally data intensive . (not data intensive)	F
15.	AHP is suitable for qualitative and quantitative information	T
16.	In a Decision Support System, the primary focus is often on the computerized components of the system	T
17.	Information systems capture data from the organization (internal data) and its environment (external data).	T
18.	Most of the data captured by information systems relates to the operations of the organization itself	T
19.	The <i>fewer resources</i> a system consumes in producing given outputs, the more effective it is. (efficient)	F
20.	The objective of management reporting systems is to provide routine information to managers	T
21.	The decision support system is under user control, from early beginning to final putting into use and daily use	T
22.	DSS refers to a class of systems, which supports the process of decision-making and does not always give a decision itself	T
23.	To increase the production level by 40 percent using the backward sensitivity analysis, first, the target value for the production level can be set	T
24.	An Inter-Organizational DSS provides stakeholders with access to a company's intranet and authority or privileges to use specific DSS capabilities	T
25.	ERP is <u>NOT</u> a Decision Support System	T
26.	In the payoff table, the rows correspond to the possible decision alternatives	T
27.	In pro forma financial model Costs are estimated based on past data, gross sales are predicted, and profit or loss is then calculated on these relationships	T
28.	DSS with simulation models conduct experiments to identify conditions or situations that approximate the actual conditions	T
29.	The passive data dictionary (also called integrated data dictionary) is the one used only for documentation purposes. (non-integrated)	F



PRACTICAL BANK (MCQ)

1.	A _____ is an interactive computer-based system or subsystem intended to help decision makers use communications technologies, data, documents, knowledge and/or models to <u>identify and solve problems</u> , complete decision process tasks , and make decisions.			
	a) DSS	b) AI	c) ML	d) CSP
2.	_____ decision-making is concerned with long-term goals & policies for resource allocation/management to meet defined objectives			
	a) Strategic	b) Management (Tactical)	c) Operational	d) None
3.	_____ decision-making is concerned with the acquisition & efficient utilization of resources to achieve defined goals			
	a) Strategic	b) Management (Tactical)	c) Operational	d) None
4.	_____ decision-making is concerned with the effective & efficient use of resources for execution of specific tasks			
	a) Strategic	b) Management (Tactical)	c) Operational	d) None
5.	Often called "programmed decisions" because they are routine and there are usually specific policies, procedures, or actions that can be identified to help make the decision			
	a) Structured	b) Semi-structured	c) Unstructured	d) None
6.	"This is how we usually solve this type of problem" means that the decision is _____.			
	a) Structured	b) Semi-structured	c) Unstructured	d) None
7.	Decision scenarios that often involve new or unique problems and the individual has little or no programmatic or routine procedure for addressing the problem or making a decision.			
	a) Structured	b) Semi-structured	c) Unstructured	d) None
8.	Decision scenarios that have some structured components and some unstructured components.			
	a) Structured	b) Semi-structured	c) Unstructured	d) None
9.	_____ analysis is the process of discovering how much change in <i>output</i> according to change in one input variable.			
	a) Sensitivity	b) Goal seeking	c) What-If	d) Model
10.	_____ analysis is the ability to determine what inputs are required to reach specific goals .			
	a) Sensitivity	b) Goal seeking	c) What-If	d) Model
11.	The process of <i>changing different set of values</i> in one or more formulas to <u>explore all various results</u> .			
	a) Sensitivity	b) Goal seeking	c) What-If	d) Model
12.	A _____ is a way of <u>working, organizing, or doing something</u> which follows a fixed plan or set of rules.			
	a) System	b) Model	c) View	d) Strategic
13.	_____ is an organized group of related objects or components.			
	a) System	b) Model	c) View	d) Strategic



14.	_____ is a computer program that describes the mechanics of the considered system.			
	a) System	b) Model	c) View	d) Strategic
15.	_____ can be used for <u>understanding and predicting the behavior</u> of systems			
	a) System	b) Model	c) View	d) Strategic
16.	_____ are computer-based programs that <u>emulate the decision-making abilities of a human expert in a particular domain</u> .			
	a) Expert systems (ES)	b) Case-based reasoning (CBR)	c) DSS	d) Unintelligent systems
17.	_____ are designed to <u>replicate</u> the problem-solving and reasoning processes that a human expert uses to make decisions in a specific area of expertise.			
	a) Expert systems (ES)	b) Case-based reasoning (CBR)	c) DSS	d) Unintelligent systems
18.	_____ utilize knowledge from human experts , often <u>stored in a knowledge base</u> , and employ an <u>inference engine to draw conclusions</u> , make decisions, or provide recommendations based on that knowledge.			
	a) Expert systems (ES)	b) Case-based reasoning (CBR)	c) DSS	d) Unintelligent systems
19.	_____ is a repository that contains information, rules, facts, and heuristics related to the specific domain of expertise. It is created by <u>capturing</u> the expertise of human specialists.			
	a) Knowledge Base	b) Inference Engine	c) User Interface	d) None
20.	_____ component is responsible for processing information from the knowledge base and applying reasoning mechanisms to draw conclusions or make decisions. It <u>simulates the decision-making process</u> of a human expert.			
	a) Knowledge Base	b) Inference Engine	c) User Interface	d) None
21.	Expert systems often include a _____ through which users can interact with the system, input information, and receive recommendations or solutions.			
	a) Knowledge Base	b) Inference Engine	c) User Interface	d) None
22.	Types of inference engines are:			
	a) Forward chaining	b) Backward chaining	c) Both A & B	d) None
23.	In a _____ engine, The <u>system starts with the available data</u> and iteratively applies rules to derive new conclusions or reach a goal. It is often used in situations where the system needs to <u>find a solution by building up from known facts</u> .			
	a) Forward chaining	b) Backward chaining	c) Both A & B	d) None
24.	_____ starts with a <u>goal or desired outcome</u> and works backward through the rules to find the supporting evidence or facts. This type of inference engine is useful in scenarios where the system needs to <u>determine the root cause of a problem</u> .			
	a) Forward chaining	b) Backward chaining	c) Both A & B	d) None
25.	_____ is a problem-solving methodology that emulates the way humans learn and make decisions based on past experiences.			
	a) Expert systems (ES)	b) Case-based reasoning (CBR)	c) DSS	d) Unintelligent systems



26.	<u>Instead of relying on predefined rules or explicit programming</u> , _____ solves new problems by recalling and adapting solutions from past experiences.						
a)	Expert systems (ES)	b)	Case-based reasoning (CBR)	c)	DSS	d)	Unintelligent systems
27.	Retrieval is the _____ step in CBR process. (Retrieval -> Reuse -> Revise -> Retain)						
a)	First	b)	Second	c)	Third	d)	Fourth
28.	_____ involves identifying cases from the case base that are similar to the current problem . Similarity is typically determined based on features or attributes relevant to the problem.						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
29.	_____ is finding similar cases from the case base .						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
30.	Once similar cases are retrieved, the next step is to apply the solutions or knowledge from those cases to the current problem. This leverages the idea that similar problems often have similar solutions.						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
31.	_____ is applying solutions from retrieved cases to the current problem.						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
32.	In _____, The reused solution may not be a perfect fit for the current problem. The system needs to adapt or revise the solution to better match the specific context and requirements of the new problem.						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
33.	_____ is modifying the solution based on the current problem context.						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
34.	After solving the current problem, the adapted case is stored back into the case base . This enriches the repository for <u>future problem-solving</u> , contributing to the learning aspect of CBR.						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain
35.	_____ is Incorporating the newly solved case into the case base for <u>future use</u> .						
a)	Retrieval	b)	Reuse	c)	Revise	d)	Retain



PRACTICAL WRITTEN QUESTIONS

Cost profit, PERT/CPM, Pricing, Forward/Backward chaining

1- Cost-profit

الطلبات:

(اكيد مش هيطلب كل حاجة 😊)

A. Optimistic, Pessimistic, and Regret Approaches

B. Expected Value Approach and EVPI

C. Decision Tree

المطلوب الاول A : عاوز **Optimistic, Pessimistic, and Regret Approaches**

“ Optimistic

“ Conservative
(Pessimistic)

“ Regret

	Profit	Cost
Optimistic	Max → max (maximax) (max of max)	Min → min (minimin) (min of min)
Conservative (pessimistic)	Min → max (maximin) (max of min)	Max → min (minimax) (min of max)
Regret	Max → max → min (min of max of max)	Min → max → min (min of max of min)

في البداية المسائل دي نوعين اما **Profit ربح** أو **Cost تكلفة** ويحدد في رأس السؤال

A.1 لاحظ ان المسألة Profit

A. According to the following profit payoff table, using the optimistic, pessimistic and regret approaches, which solution decision maker choose form A,B, and C

B. If decision maker know that the probabilities of $s_1=0.2$, $s_2=0.3$, and $s_3=0.5$, which decision according to expected value approach, and the value of EVPI?

	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16

دي المسألة اللي هنحلها



A.1.1 Optimistic Approach (Maximax) ($\max \rightarrow \max$):

- Identify the **maximum payoff** for each alternative:

هنختار هنا أكبر القيم في A, B, C

A: Payoffs = 4, -4, 9 $\rightarrow$ Max = 9.

B: Payoffs = 5, 8, 3 $\rightarrow$ Max = 8.

C: Payoffs = 6, 12, 16 $\rightarrow$ Max = **16**.

- Choose the alternative with the **maximum of these values**:

وهنختار أكبر قيمة فيهم الثلاثة

- Decision: C.**

A.1.2 Pessimistic Approach (Maximin) ($\min \rightarrow \max$):

- Identify the **minimum payoff** for each alternative:

هنختار هنا أصغر القيم في A, B, C

- A:** Payoffs = 4, -4, 9 $\rightarrow$ Min = -4.

- B:** Payoffs = 5, 8, 3 $\rightarrow$ Min = 3.

- C:** Payoffs = 6, 12, 16 $\rightarrow$ Min = 6.

- Choose the alternative with the **maximum of these minimums**:

وهنختار أكبر قيمة فيهم الثلاثة

- Decision: C.**

A.1.3 Regret Approach (Minimax Regret) (max → max → min):

- 1-هنتار أكبر رقم في كل state (عمود)
- 2-بعد كذا هنتارح (الرقم اللي اخترناه من S1 - قيم العمود)
- 3-هنتار أكبر القيم في كل "صف"
- 4- وفي الآخر هنتار أقل قيمه فيهم

a. Construct the Regret Table:

1. Determine the **best payoff for each state (S1, S2, S3):**

- **S1:** Best payoff = 6
- **S2:** Best payoff = 12
- **S3:** Best payoff = 16

2. Calculate the regret using the formula:

Regret = Max payoff in state – Payoff of the alternative (من الجدول)

1. Regret Table:

Alternative	S1	S2	S3
A	$6-4=2$	$12-(-4)=16$	$16-9=7$
B	$6-5=1$	$12-8=4$	$16-3=13$
C	$6-6=0$	$12-12=0$	$16-16=0$

Identify the **maximum regret** for each alternative:

خلي بالك "أكبر قيمه في كل صف"

A: Max regret = 16

B: Max regret = 13

C: Max regret = 0

- ♦ Choose the alternative with the ****minimum** of the maximum regrets**:

- **Decision: C.**

A.2 في حالة ان المسألة كانت Cost:

A.2.1 Optimistic Approach (Minimin) ($\min \rightarrow \min$):

- Identify the **minimum cost** for each alternative:

هنختار هنا أصغر القيم في A, B, C

- A: Costs = 4, -4, 9 $\rightarrow$ Min = -4.
- B: Costs = 5, 8, 3 $\rightarrow$ Min = 3.
- C: Costs = 6, 12, 16 $\rightarrow$ Min = 6.

- Choose the alternative with the **minimum of these values**:

وهنختار أصغر قيمة فيهم الثلاثة

- Decision: A.

A.2.2 Pessimistic Approach (Minimax) ($\max \rightarrow \min$):

- Identify the **maximum cost** for each alternative:

هنختار هنا أكبر القيم في A, B, C

- A: Costs = 4, -4, 9 $\rightarrow$ Max = 9.
- B: Costs = 5, 8, 3 $\rightarrow$ Max = 8.
- C: Costs = 6, 12, 16 $\rightarrow$ Max = 16.

- Choose the alternative with the **minimum of these maximums**:

وهنختار أصغر قيمة فيهم الثلاثة

- Decision: B.

A.2.3 Regret Approach (Minimax Regret) (min → max → min):

هنا بقا : 

- 1- هنتار أقل رقم في كل state (عمود)
- 2- بعد كذا هنتارح (قيم العمود - الرقم اللي اخترناه من S1) 3- هنتار أكبر القيم في كل "صف"
- 4- وفي الآخر هنتار أقل قيمه فيهم

1. Determine the **minimum cost for each state (S1, S2, S3):**

خلي بالك "أصغر قيمة في كل عمود" 

- **S1:** Min cost = 4
- **S2:** Min cost = -4
- **S3:** Min cost = 3

2. Calculate the regret using the formula:

Regret = Payoff of the alternative (من الجدول) – Min payoff in state

Alternative	S1	S2	S3
A	4-4= 0	-4-(-4)= 0	9-3= 6
B	5-4= 1	8-(-4)= 12	3-3= 0
C	6-4= 2	12-(-4)= 16	16-3= 13

Identify the **maximum regret** for each alternative:

- **A:** Max regret = 6.
- **B:** Max regret = 12.
- **C:** Max regret = 16.

خلي بالك "أكبر قيمة في كل صف" 

Choose the alternative with the **minimum of these maximum regrets:**

وهنتار أصغر قيمة فيهم الثلاثة

- **Decision: A.**

المطلوب الثاني: عايز قرار بناء على ال **expected value approach** وقيمة ال **EVPI**

B.1 Expected Value for Each Alternative:

Given probabilities:

$$P(S1) = 0.2$$

$$P(S2) = 0.3$$

$$P(S3) = 0.5$$

Use the formula:

$$\text{Expected Value (EV)} = \sum (\text{Payoff} \times \text{Probability})$$

Alternative	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16
probabilities	p(S1)= 0.2	P(S2)=0.3	P(S3)=0.5

♦ A:

$$EV A = (4 \times 0.2) + (-4 \times 0.3) + (9 \times 0.5) = 0.8 - 1.2 + 4.5 = 4.1$$

♦ B:

$$EV B = (5 \times 0.2) + (8 \times 0.3) + (3 \times 0.5) = 1 + 2.4 + 1.5 = 4.9$$

♦ C:

$$EV C = (6 \times 0.2) + (12 \times 0.3) + (16 \times 0.5) = 1.2 + 3.6 + 8 = 12.8$$

في حالة ان المسألة Profit هنختار الاكبر

♦ **Decision: C** (since EV C=12.8 is the **highest**).

في حالة ان المسألة Cost هنختار الاصغر

♦ **Decision: A** (since EV A=4.1 is the **Lowest**).

B.2 Expected Value with Perfect Information (EVPI):

1. For each state, multiply the **best payoff** by its probability:

هنختار اكبر رقم في كل state و هنضربه في الاحتمال بتاعه 🚩

- S1 : $6 \times 0.2 = 1.2$

- S2 : $12 \times 0.3 = 3.6$

- S3 : $16 \times 0.5 = 8$

2. Calculate EVPI:

$$EVPI = 1.2 + 3.6 + 8 = 12.8$$

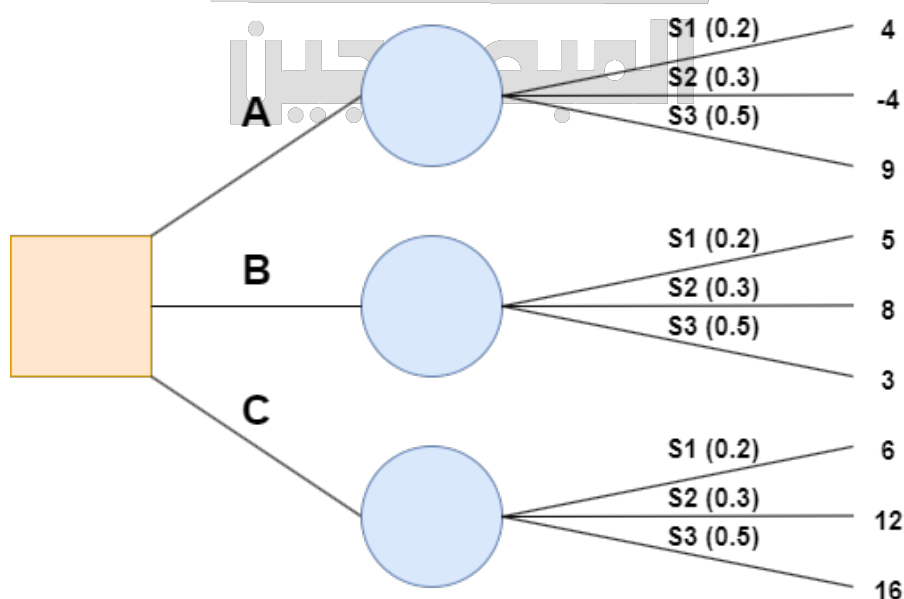
B.3 Calculate the Value of Perfect Information (EVPI):

$$VPI = EVPI - \text{Highest Expected Value} = 12.8 - 12.8 = 0$$

المطلوب الثالث: Decision Tree

بنرسم مربع طالع منه ال alternatives وبعدين بنطلع دواير طالعه منها ال states وبتكتب جنبها الاحتمالات وفي الآخر خالص بتكتب ال payoffs في طرف الخطوط

	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16



2- PERT/CPM

ES	Duration	EE
Task Name		
LS	slack	LE

• Duration معطي في السؤال

• خلي بالك من كل Activity مين هو dependency بتاعه

• المسألة مكونه من forward pass و backward pass

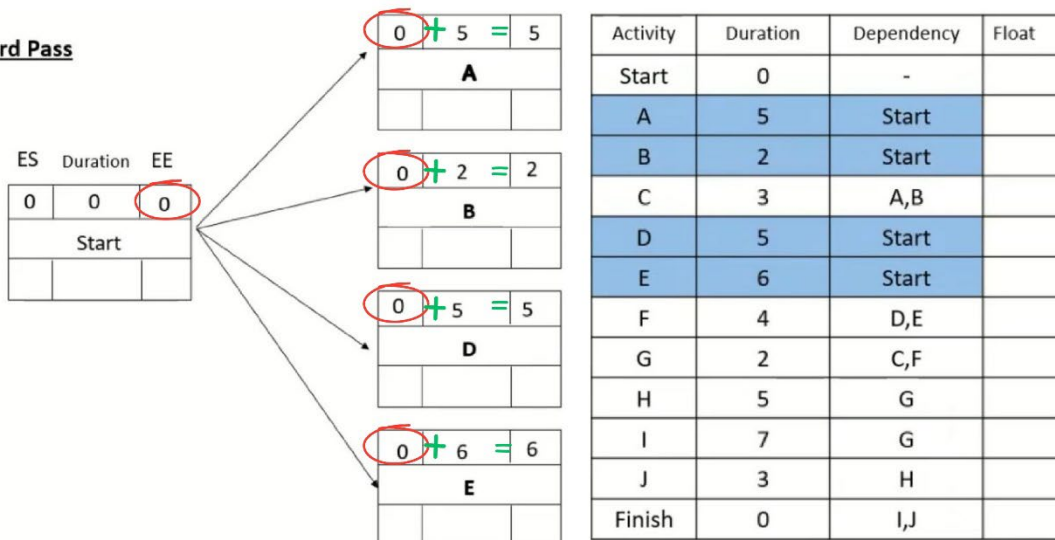
Exercise on Critical Path:

For the following table of information,

1. Draw the network diagram
2. Determine the slack for each activity
3. List the network paths
4. Determine the critical path(s)

Activity	Duration	Dependency	Slack
Start	0	-	
A	5	Start	
B	2	Start	
C	3	A,B	
D	5	Start	
E	6	Start	
F	4	D,E	
G	2	C,F	
H	5	G	
I	7	G	
J	3	H	
Finish	0	I,J	

1. forward Pass



• لاحظ ان start كل حاجه فيها ب صفر

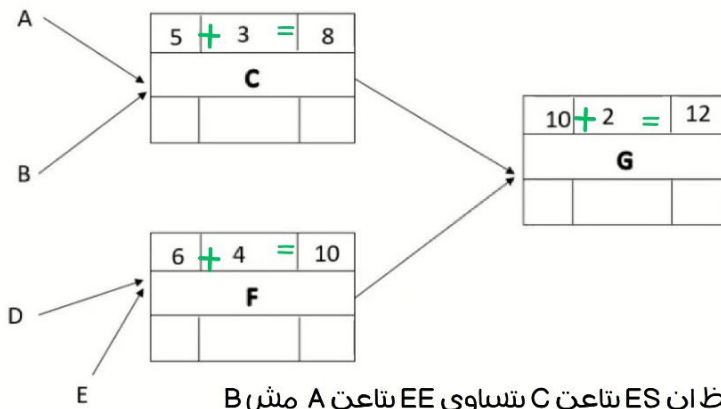
• لاحظ ان A, B, D, E كلهم ال Dependency بتاعهم هو start "اللي بالازرق"

• ال EE بتاع dependency هو ES بتاع Activity اللي معتمد عليه "اللي بالاحمر"

• ولو فيه Activity له اتنين dependency بنأخذ الاكبر

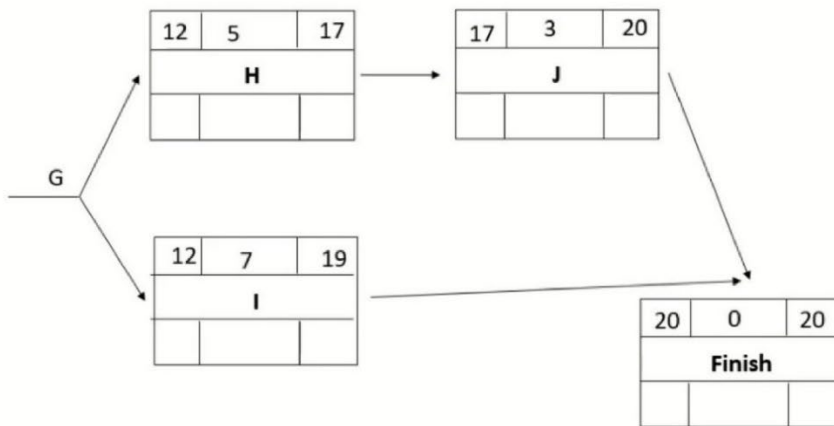
• لاحظ ان $EE = Duration + ES$ "بالاخضر"





• لاحظ ان ES بتاعت C بتساوي EE بتاعت A مش B
عشان A اكبر وكذلك في G هتلاقي F اكبر

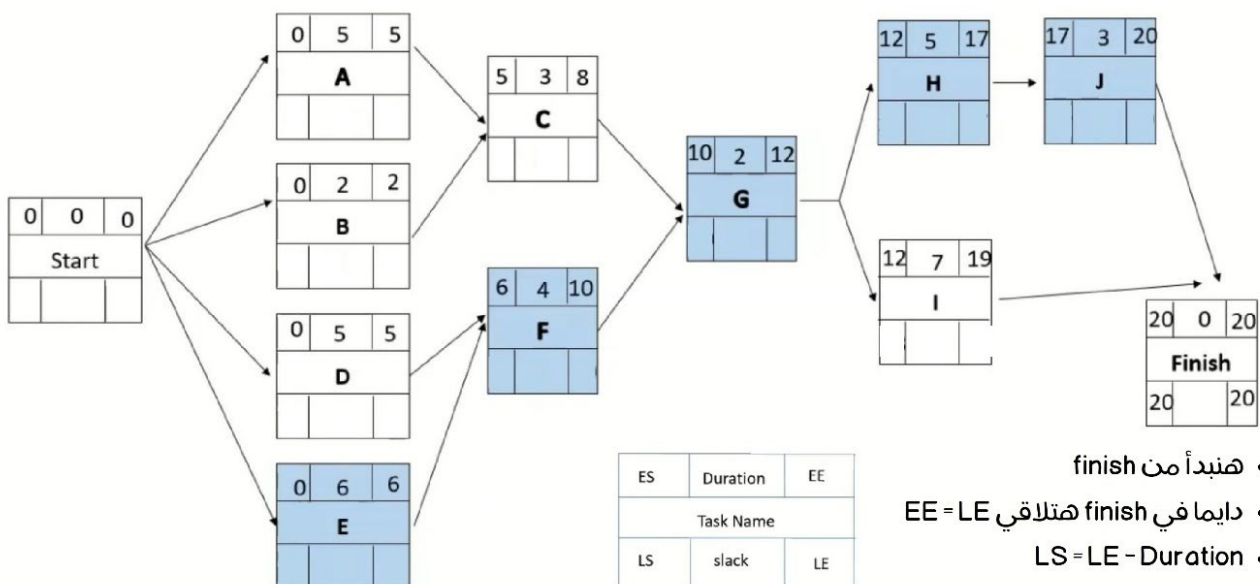
Activity	Duration	Dependency	Slack
Start	0	-	
A	5	Start	
B	2	Start	
C	3	A,B	
D	5	Start	
E	6	Start	
F	4	D,E	
G	2	C,F	
H	5	G	
I	7	G	
J	3	H	
Finish	0	I,J	



• كذا خلصنا forward pass

Activity	Duration	Dependency	Float
Start	0	-	
A	5	Start	
B	2	Start	
C	3	A,B	
D	5	Start	
E	6	Start	
F	4	D,E	
G	2	C,F	
H	5	G	
I	7	G	
J	3	H	
Finish	0	I,J	

backward pass

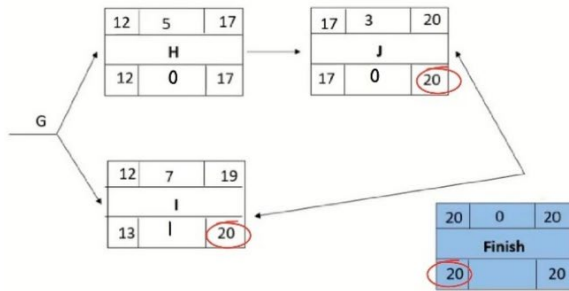


• هنبداً من finish

• دايماً في finish هتلاقي $EE = LE$

• $LS = LE - \text{Duration}$

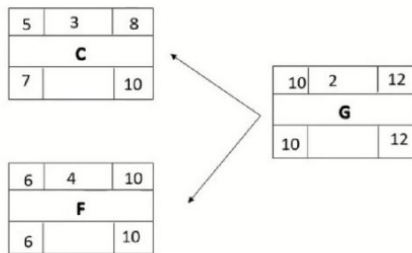




- لاحظ ان LS بتاعت Activity (في الحالة دي هي finish) بتساوي LE بتاعت dependency (اللي هي I, J) - بالاحمر

$$LS = LE - \text{Duration}$$

• ملحوظة: القانون دا في Task الواحدة مثمن بين كذا Task (يعني LS بتاعت I بتساوي LE بتاعت I - ال Duration بتاعت I)

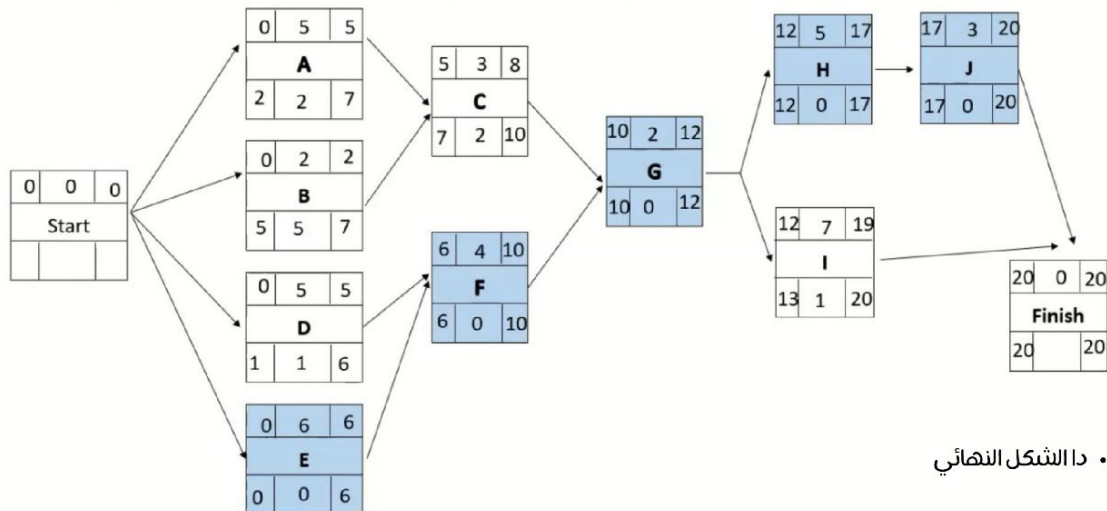


$$\text{slack} = LE - EE \text{ او } LS - ES$$

• ملحوظة: القانون دا في Task الواحدة مثمن بين كذا Task

- خلي بالك وانت رايح من H و I الى G هتلاقى عندك اثنين LS <= خذ الاقل دايمًا يكون هو LE بتاعت G

Network Diagram & The slack for each activity



• دا الشكل النهائي

أسهل طريقة تجيب بيها ال **critical path** هي انك تشوف في الرسة التاسكات اللي عندها **slack** ب0 زي الرسة اللي فوق
• عشان تحسب Critical path اكبر مسار من حيث Duration

Network paths:

1. A - C - G - H - J = 18
2. A - C - G - I = 17
3. B - C - G - H - J = 15
4. B - C - G - I = 14
5. D - F - G - H - J = 19
6. D - F - G - I = 18
7. E - F - G - H - J = 20 <=====
8. E - F - G - I = 19

Critical path:

E - F - G - H - J



3- Pricing

What is the best price that achieves the highest profit for the next five years? provided that the increase or decrease between each year =1.

Year Pound	1	2	3	4	5
6	9	13	10	14	7
7	13	10	11	9	13
8	11	8	14	6	17
9	21	7	9	5	16



Year Pound	1	2	3	4	5
6				21	7
7				27	13
8				26	17
9				22	16

- في السؤال دا بنبدأ من الآخر
- حط نفس الأرقام اللي في العمود الأخير في المربعات الصفراء
- كل مربع هيطلع منه سهم ل 3 مربعات في العمود اللي بعده عدا الأول والآخر
- 7 مثلاً ل 4 و 9 شوف الأكبر واجمع عليه 7 وحطه في المربع الأصفر " $21 = 14 + 7$ "
- 13 ل 4 و 9 و 6 الأكبر هو 14 " $27 = 14 + 13$ "
- 17 ل 9 و 6 و 5 الأكبر هو 9 " $26 = 9 + 17$ "
- 16 ل 6 و 5 بس الأكبر هو 6 " $22 = 6 + 16$ "
- هتكرر نفس الخطوات للآخر



Year Pound	1	2	3	4	5
6	58 9	45 13	32 10	21 14	7 7
7	67 13	54 10	41 11	27 9	13 13
8	71 11	50 8	40 14	26 6	17 17
9	65 21	44 7	36 9	22 5	16 16

- دا شكل الجدول في النهاية
- ابدأ من العمود الاول . شوف اكبر قيمه في المربع الاصفر "7"
- ونفس الخطوات هتعملها ثاني
- 7 ل 54 و 50 و 44 هتأخذ الاكبر وهكذا

ANSWER:



Year Pound	1	2	3	4	5
6	58 9	45 13	32 10	21 14	7 7
7	67 13	54 10	41 11	27 9	13 13
8	71 11	50 8	40 14	26 6	17 17
9	65 21	44 7	36 9	22 5	16 16

Years	1	2	3	4	5
Pound	8	7	7	7	8
Profit	71 +	54 +	41 +	27 +	17 = 210

Total Profit = 210 pounds



4- Forward/Backward chaining

Forward:

A , B , C , D , E
X , L , Y , Z

في المسألة يكون فيه facts (معطى في السؤال) و Rules (معطى في السؤال)

عدي على كل ال rules شوف تقدر تحققها ولا لا

R1: $Y \wedge D \rightarrow Z$

R2: $X \wedge B \wedge E \rightarrow Y$

R3: $A \rightarrow X$

R4: $C \rightarrow L$

R5: $L \wedge M \rightarrow N$

forward chaining

C1: X, L

C2: Y

C3: Z

دا الحل اللي هتكتبه

C1 {

R1 مئش معايا Y
R2 مئش معايا X
R3 مئش معايا A ضيف الناتج "X" مع facts
R4 مئش معايا C ضيف الناتج "L" مع facts
R5 مئش معايا M

C2 {

R1 مئش معايا Y
R2 مئش معايا X, B, E ضيف الناتج "Y" مع facts
R3 خلصتها
R4 خلصتها
R5 مئش معايا M

C3 {

R1 مئش معايا Y, D ضيف الناتج "Z" مع facts
R2 خلصتها
R3 خلصتها
R4 خلصتها
R5 مئش معايا M

تمام كدا

المفروض انه بيعطيك goal توصله عشان متعلم مئش كلهم

Backward:

A , B , C , D , E

goal: Z

R1: $Y \wedge D \rightarrow Z$

R2: $X \wedge B \wedge E \rightarrow Y$

R3: $A \rightarrow X$

R4: $C \rightarrow L$

R5: $L \wedge M \rightarrow N$

backward chaining

goal $\rightarrow$ z

subgoal (1) $\rightarrow$ y

subgoal (2) $\rightarrow$ x

fire X

fire Y

fire Z

المفروض اني اوصل لل goal بالعكس

ال goal في R1 بس مئش معايا Y $\rightarrow$ goal

Y في R2 بس مئش معايا X $\rightarrow$ subgoal

X في R3 $\rightarrow$ subgoal

